Water-Based Proof of Jensen's Inequality

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Abstract. Various proofs of Jensen's inequality exist. This note introduces a novel proof using water and potential energy concepts. By modeling the inequality using water, we establish a physical analogy that demonstrates the theorem visually and intuitively. This work is a generalization of [1] and [2].

1. PHYSICAL PROOF OF JENSEN'S INEQUALITY. Jensen's inequality applies to all convex functions. However, because this proof uses a physical analogy, we require the stronger condition that the function's second derivative is continuous and positive.

Theorem 1 (Jensen's inequality). *Let $f : \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function such that $f''(x) > 0$ for all real numbers x. Given real numbers $x_1, x_2, \cdots, x_n$ and positive weights $p_1, p_2, \cdots, p_n$ such that $\sum_{k=1}^{n} p_k = 1$, the inequality is given by:*

$$\sum_{k=1}^{n} p_k f(x_k) \geq f(\sum_{k=1}^{n} p_k x_k). \tag{1}$$

Water-based proof of Jensen's inequality. For any real numbers a, b, c, define the function

$$g(x) = f(a + x) + bx + c. \tag{2}$$

Then the following statements

$$f''(a + x) > 0 \Leftrightarrow g''(x) > 0, \tag{3}$$

$$\sum_{k=1}^{n} p_k f(a + x_k) \geq f(\sum_{k=1}^{n} p_k (a + x_k)) \Leftrightarrow \sum_{k=1}^{n} p_k g(x_k) \geq g(\sum_{k=1}^{n} p_k x_k), \tag{4}$$

are true. Thus, we may assume $x_1, x_2, \cdots x_n$ are all positive numbers, $f(0) = 0$, and $f'(0) = 0$ without loss of generality.

Consider a massless container with gravitational potential energy $f(V)$ when it holds a volume V of water. (The method of constructing such a container will be explained later.) Let $A(h)$ be the cross-sectional area of this container at height h (see Figure 1).

Now let's prepare n water containers. The cross-sectional area at height h of the k-th container is $p_k A(h)$. Arrange the containers in a line, connect them with hoses, and seal the gaps between them tightly so water cannot pass through. Then, fill the k-th container with a volume $V_k = p_k x_k$. This setup is equivalent to having the container in Figure 1 filled with a volume of x_k and then scaling the cross-sectional area by p_k.

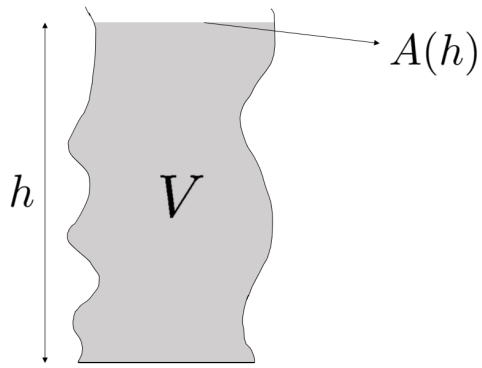

Figure 1. Container with potential energy $f(V)$ corresponding to a water volume V.

Thus, the potential energy of water in the k-th container will be $p_k f(x_k)$ (see Figure 2). Therefore, the initial gravitational potential energy is

$$U_i = \sum_{k=1}^{n} p_k f(x_k). \tag{5}$$

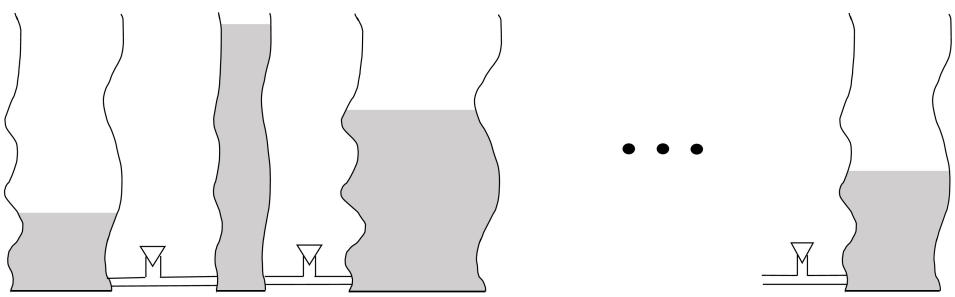

Figure 2. The initial state of n containers.

When the hoses are opened, allowing the water to flow freely, the water levels fluctuate until they stabilize at equilibrium. Once equilibrium is reached, there is no net flow between the containers, meaning there is no pressure difference at the ends of any connecting hose. Because pressure is an injective function dependent solely on depth, the lack of a pressure difference indicates that all the containers have the same water level (see Figure 3).

In equilibrium, the entire system acts as a single container with cross-sectional area $A(h)$, holding a total water volume $V_{total} = \sum_{k=1}^{n} p_k x_k$. Thus, the gravitational potential energy at equilibrium is

$$U_{eq} = f(\sum_{k=1}^{n} p_k x_k). \tag{6}$$

Since energy is lost when the water sloshes, the following inequality holds:

$$U_i \geq U_{eq}. \tag{7}$$

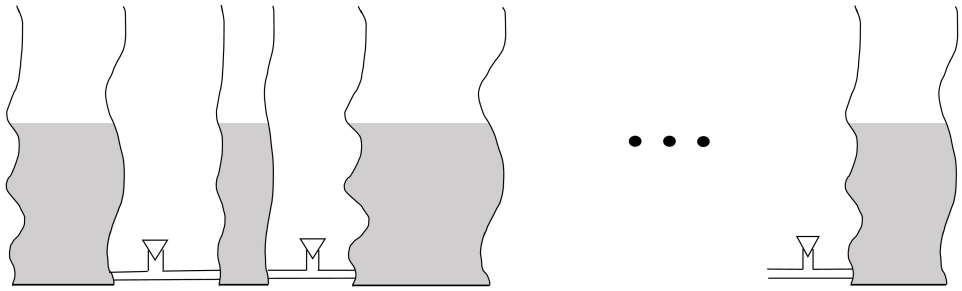

Figure 3. The equilibrium state of n containers.

(A more detailed justification for (7) is provided below.) Therefore, Jensen's inequality holds:

$$\sum_{k=1}^{n} p_k f(x_k) \geq f(\sum_{k=1}^{n} p_k x_k). \tag{8}$$

■

2. REMAINING ISSUES.

Why $U_i \geq U_{eq}$ holds. Mathematicians seeking rigor might find the 'physical reasoning' behind (7) unsatisfactory, as I did not rigorously prove why the system eventually reaches equilibrium. Below is a rigorous proof of why (7) holds.

Denote the height of the water level of the k-th container in the initial state as h_k ($k = 1, 2, \cdots n$), and denote the height of the water level in the equilibrium state as h_{eq}. For each $k = 1, 2, \cdots n$, Color the region $0 \leq h \leq \min(h_k, h_{eq})$ in the k-th container with light gray. If $h_k \geq h_{eq}$, color the region $h_{eq} \leq h \leq h_k$ in the k-th container with dark gray. If $h_k \leq h_{eq}$, color the region $h_k \leq h \leq h_{eq}$ in the k-th container with black. (See Figure 4)

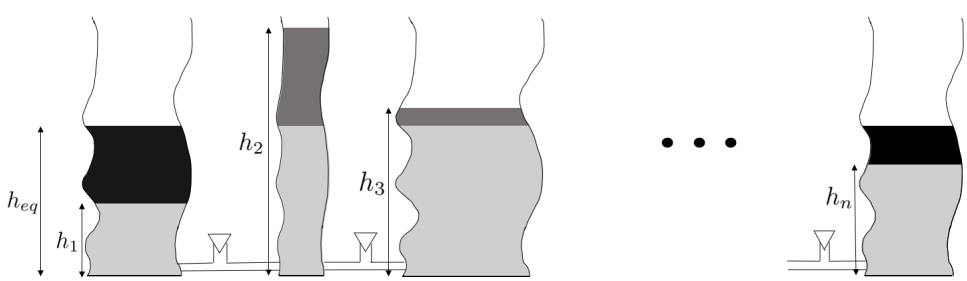

Figure 4. Regions colored.

Denote the region colored in light gray as region A, the region colored in dark gray as region B, and the region colored in black as region C. Let V_A, V_B, and V_C represent the volumes of regions A, B, and C, respectively. Let h_A, h_B, and h_C represent the heights of the centers of mass of regions A, B, and C, respectively. Since the volume of water is conserved, we have:

$$V_A + V_B = V_A + V_C. \tag{9}$$

Thus,

$$V_B = V_C. \tag{10}$$

It is obvious that the following inequality holds:

$$h_C \leq h_{eq} \leq h_B. \tag{11}$$

We can also express U_i and U_{eq} with these notations as follows:

$$U_i = V_A h_A + V_B h_B, \quad U_{eq} = V_A h_A + V_C h_C. \tag{12}$$

(For simplicity, we assume the density of water and the gravitational acceleration are 1.) By equations (10) and (11),

$$\begin{aligned} U_{eq} - U_i &= (V_A h_A + V_C h_C) - (V_A h_A + V_B h_B) \\ &= V_C h_C - V_B h_B \\ &= V_C(h_C - h_B) \quad \text{by (10)} \\ &\leq 0 \quad \text{by (11).} \end{aligned}$$

This provides a rigorous proof of the inequality

$$U_i \geq U_{eq}. \tag{13}$$

How to construct such a container for *any* given C^2 function f with a positive second derivative. To construct a container with a gravitational potential energy $f(V)$ when it holds a volume V of water, we need to ensure that the cross-sectional area is positive. Suppose such a container exists.

Adding a small volume dV of water to a container filled to height h increases the gravitational potential energy by $h \cdot dV$. Thus, we get:

$$f'(V) = \frac{df(V)}{dV} = h. \tag{14}$$

If we express volume V as a function of h, this gives:

$$V(h) = (f')^{-1}(h). \tag{15}$$

(Note that $(f')^{-1}$ is the inverse function of f' and is not to be confused with $1/f'$.) Since f' is a strictly increasing continuous function, the image of f' is an open interval, and $(f')^{-1}$ is well defined in that interval.

Next, we derive the cross-sectional area $A(h)$:

$$A(h) = V'(h) = ((f')^{-1})'(h) = \frac{1}{f''((f')^{-1}(h))} > 0. \tag{16}$$

The positivity of the second derivative ensures that the cross-sectional area is positive and well-defined. With the assumptions $f(0) = 0$ and $f'(0) = 0$ already made, we observe that a container with a cross-sectional area $A(h) = ((f')^{-1})'(h)$ has gravitational potential energy $f(V)$ when it holds a volume V of water, as determined by simple integral calculations.

When $A(h) = 1$, we have $f(V) = \frac{1}{2}V^2$. Using Jensen's inequality for x^2, we can derive the Cauchy-Schwarz inequality, thereby validating [1]. When $A(h) = (p-1)h^{p-2}$, we have $f(V) = \frac{1}{q}V^q$. Using Jensen's inequality for x^q, we can prove Hölder's inequality, thereby validating [2].

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